

Mobius 3 L Single-use Bioreactor - 3 L vessel

merck-mobius-3

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	1 L	Source-supported
Maximum working volume	2.4 L	Source-supported
Nominal vessel / bag size	3 L	Source-supported
Total vessel volume loaded into the form	3 L	Source-supported
Initial working volume loaded into the form	1 L	Derived / estimated
Product-contact material	Polycarbonate vessel/shaft; HDPE headplate/impeller; silicone seals	Source-supported
Height : diameter	1.24	Derived / estimated
Internal / chamber diameter	0.135 m	Source-supported
Liquid height at maximum working volume	0.1677 m	Derived / estimated
Impeller / mixing type	Marine / placeholder mixing element	Source-supported
Impeller count	1	App assumption
Impeller diameter	0.0762 m	Source-supported
Power number	0.3	Source-supported
Initial agitation	100 rpm	App assumption
Maximum agitation	250 rpm	App assumption
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.3 vvm	App assumption
Baffled vessel	No	App assumption
Maximum pressure	0.5 bar(g)	App assumption
Default operating pressure	0.05 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 1 L

Source-supported | Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2; Specifications - Volume, Dimensions, Materials

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1 L.

Limitations: May 2023 dedicated specification, now archived locally. Overall height 244 mm is NOT liquid height. Working range is 1-2.4 L, distinct from nominal 3 L capacity.

Local original: Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2

Manufacturer publication

Maximum working volume - 2.4 L

Source-supported | Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2; Specifications - Volume, Dimensions, Materials

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2.4 L.

Limitations: May 2023 dedicated specification, now archived locally. Overall height 244 mm is NOT liquid height. Working range is 1-2.4 L, distinct from nominal 3 L capacity.

Local original: Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2

Manufacturer publication

Nominal vessel / bag size - 3 L

Source-supported | Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2; Specifications - Volume, Dimensions, Materials

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 3 L.

Limitations: May 2023 dedicated specification, now archived locally. Overall height 244 mm is NOT liquid height. Working range is 1-2.4 L, distinct from nominal 3 L capacity.

Local original: Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2

Manufacturer publication

Total vessel volume loaded into the form - 3 L

Source-supported | Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2; Specifications - Volume, Dimensions, Materials

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 3 L.

Limitations: May 2023 dedicated specification, now archived locally. Overall height 244 mm is NOT liquid height. Working range is 1-2.4 L, distinct from nominal 3 L capacity.

Local original: Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2

Manufacturer publication

Initial working volume loaded into the form - 1 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(1 L, 2.4 L) = 1 L

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Polycarbonate vessel/shaft; HDPE headplate/impeller; silicone seals

Source-supported | Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2; Specifications - Volume, Dimensions, Materials

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Polycarbonate vessel/shaft; HDPE headplate/impeller; silicone seals.

Limitations: May 2023 dedicated specification, now archived locally. Overall height 244 mm is NOT liquid height. Working range is 1-2.4 L, distinct from nominal 3 L capacity.

Local original: Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2

Manufacturer publication

Height : diameter - 1.24

Derived / estimated | Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2; Specifications - Volume, Dimensions, Materials

Rule: $H = 4 V / (\pi D^2)$; liquid $H/D = H / D$

Selection / calculation: $V = 0.0024 \text{ m}^3$; $D = 0.135 \text{ m}$; $H = 0.16766940506800493 \text{ m}$; $H/D = 1.2419955930963327$

Limitations: Derived liquid aspect ratio. Does not reuse an overall vessel ratio or generic profile ratio as if it described the liquid.

Local original: Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2

Manufacturer publication

Internal / chamber diameter - 0.135 m

Source-supported | Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2; Specifications - Volume, Dimensions, Materials

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.135 m.

Limitations: May 2023 dedicated specification, now archived locally. Overall height 244 mm is NOT liquid height. Working range is 1-2.4 L, distinct from nominal 3 L capacity. Conflict retained: AN12431EN (July 2023), Table 1, gives 137 mm; its cited SP2345000 Rev. B (August 2010) also explicitly gives 137 mm inner diameter. Selected 135 mm from the newer dedicated May 2023 datasheet. No manufacturer correction or revision explanation found; not resolved by averaging.

Local original: Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2

Manufacturer publication

Liquid height at maximum working volume - 0.1677 m

Derived / estimated | Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2; Specifications - Volume, Dimensions, Materials

Rule: $H = 4 V / (\pi D^2)$; liquid $H/D = H / D$

Selection / calculation: $V = 0.0024 \text{ m}^3$; $D = 0.135 \text{ m}$; $H = 0.16766940506800493 \text{ m}$; $H/D = 1.2419955930963327$

Limitations: Ideal-cylinder estimate from published diameter and working volume. Ignores tapered walls, curved bottoms and internals; not a manufacturer liquid-height specification.

Local original: Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2

Manufacturer publication

Impeller / mixing type - Marine / placeholder mixing element

Source-supported | Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2; Specifications - Volume, Dimensions, Materials

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Marine / placeholder mixing element.

Limitations: May 2023 dedicated specification, now archived locally. Overall height 244 mm is NOT liquid height. Working range is 1-2.4 L, distinct from nominal 3 L capacity.

Local original: Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2

Manufacturer publication

Impeller count - 1

App assumption | Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2; Table 1 and Figure 1

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 1.

Limitations: One shaft-mounted impeller is depicted/described in Table 1 and Figure 1. Replaces generic two-impeller scaling; count is a design interpretation, not an explicit numeric count specification.

Local original: [Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 \(July 2023\) - PDF page 2](#)

Impeller diameter - 0.0762 m

Source-supported | Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023) - PDF page 2; Specifications - Volume, Dimensions, Materials

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.0762 m.

Limitations: May 2023 dedicated specification, now archived locally. Overall height 244 mm is NOT liquid height. Working range is 1-2.4 L, distinct from nominal 3 L capacity.

Local original: [Merck Mobius 3 L - DS26770000 Ver. 3.0 \(May 2023\) - PDF page 2](#)

[Manufacturer publication](#)

Power number - 0.3

Source-supported | Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2; Table 1, 3 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.3.

Limitations: Published characterization value for the up-pumping, three-blade marine design; replaces generic 0.4. Not a universal value for other impellers or flow regimes.

Local original: [Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 \(July 2023\) - PDF page 2](#)

Initial agitation - 100 rpm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baseRpm = 100`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial agitation setting loaded into the form.

Maximum agitation - 250 rpm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxRpm = 250`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the ceiling available to the generic DO-control cascade.

Initial aeration - 0.05 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baseVvm = 0.05`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 0.3 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxVvm = 0.3`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - No

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baffled = false`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 0.5 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxPressure = 0.5`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0.05 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].operatingPressure = 0.05`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: `maxOxygenFraction = 100`

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

AN12431EN p. 2 supports N_p 0.3. Diameter disagreement is an edition-level conflict: 135 mm in DS26770000 (2023) versus 137 mm in SP2345000 (2010) and AN12431EN (2023). No geometric correction is claimed. AN12431EN p. 4 includes 400 rpm at 2.4 L as a performance point, not a verified universal motor limit; the RPM ceiling remains an app assumption.

Geometry caution: AN12431EN p. 2 prints minimum/maximum liquid-height ratios greater than one; PG12163EN p. 4 repeats this. A reversed ratio heading is plausible but unconfirmed. The maximum-fluid/total-height ratio 0.8 is nominal, not an absolute liquid-height measurement. Neither ratio overwrites the existing liquid-height calculation. Validation scope and conflicts: [sources/AN12431-REVIEW.md](#).

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Merck Mobius 3 L - DS26770000 Ver. 3.0 (May 2023)

[merck-mobius3-ds26770000-v3-2023-49b70bf7f7c8.pdf](#)

SHA-256: 49b70bf7f7c8574568a2fdb916fecaa410edaa81972b6310810308a78b92b58

Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023)

merck-an12431-v1-2023-9372dd222c24.pdf

SHA-256: 9372dd222c246b5b420a883bb89f5132e68dacbbea4e32164dfc0fbb48f41aeb